#67 Prof. KARL FRISTON 2.0 [Unplugged]

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in this episode of street talk unplugged so the free energy principle um originally emerged from systems neuroscience as a way a principled way of understanding what the brain does and how it does it subsequently the principles proved to be so simple and so powerful that they've been applied in a variety of contexts now so one could almost regard the free energy principle as a an organizing principle for any living system that shows the characteristics of life in this episode we'll be doing a bit of epistemic foraging with professor carl fristen legendary neuroscientist and inventor of the free energy principle the basic idea is that our brains are in the game of predicting sensory imports to try and infer the causes that generate sensations and to work out states of the world so that we can respond adaptively now long ago in my own personal epistemic journey i learned an important truth that simple and easy are not the same thing our shared human journey is filled with examples of simple ideas that were nonetheless hard to discover and some that even once explained remain hard to comprehend their subtle simplicity belies their far-reaching and deep consequences examples might include the principle of relativity quantum mechanics the principle of parsimony and entropy i think the free energy principle is another example profound and far-reaching yet belied by its simplicity it has on the one hand been dismissed as a triviality or even a tautology and on the other hailed as revolutionary and everything in between it has the potential to reshape how we view the connection between inanimate matter and living things and even to answer the question of how and why consciousness and intelligence might emerge from physical matter and processes there are of course newer attempts to explain the free energy principle such as carl's own recent paper called the free energy principle made simpler but not too simple the paper develops by starting from a world modeled as a stochastic differential equation now leaving aside the technical jargon that means a world where all things from particles to people to the largest systems all move or evolve according to two processes which combine the first is a smooth flowing evolution think of planets orbiting a star or waves rippling over water or through quantum fields the second is a random process that knocks around the smooth flow in unpredictable ways think of twinkling stars or audio static you might think of these two processes as a kind of order and chaos or yin and yang forever intertwined and enmeshed these two very different effects combine into a chaotic flow which may be entangled in a kind of tropical storm while still maintaining a semblance of structure and things think about raindrops undulating down to earth in a state of constant flux yet still droplets or a human being growing and adapting to the surprises of life all the while remaining an individual physical entity biology often asks the question what must things do in order to exist professor pristin has turned that question on its head and developed perhaps the ultimate existential formalism he asks if things exist what must they do from the foundation of stochastic differential equations tristan demonstrates that things which are defined by a markov blanket must always move towards a pullback attractor a special set of attracting states which maintains the integrity of the markov blanket and therefore a thing's coherence and identity over time one of the profound consequences of this is that the dynamics of such a system its laws of motion will manifest a form of flow dynamics which can be interpreted as bayesian active inference in other words such a thing maintains an internal equivalent of a generative model encoding beliefs about the world and itself the thing uses the model to decide actions and then performs a bayesian update based on the outcomes for me the idea that inference something widely perceived as purely abstract or mathematical the idea that it can be driven by simple laws of motion dynamically maintaining the boundaries between things maintaining order in the face of chaos is frankly astonishing what's more the free energy principle is so general that it applies at all scales of size and time leading to an ecosystem of things interacting across scales perhaps in that multi-scale active inference we might finally find the keys to a mathematics of emergence and consciousness so it's so good to see you it's uh been what about a year since since our show and i've been looking forward to chatting with you it doesn't seem like a year yes suppose it has two kids yeah about it's been almost about a year actually a little bit longer three time goes very quickly during the uh career device are you all back up to normal now or still risking kind of not not back up to normal still some restrictions and they they change very often and you know it's it's a bit confusing but you know it's more normal than it was i'll say that that's an interesting concept right by the way i just listened to your um uh kurt's theory of everything podcast right that was that the four-hour one um there was a very long one i think it was at least three hours it was the one where at the end you were talking about de-realization and de-personalization and getting into psychology well almost psychotherapy if i remember correct correctly yeah psychotherapy that's a better way to put it quite often um because we do say this about you carl that um you have a burden of knowledge and we mean this in the nicest possible way so you always know five um synonyms for a word in every single domain of science and also i mean you you even have a background in psychotherapy so it does create a tendency for conversations to potentially go anywhere and when we listened to kurt's interview of you last time um i was really surprised that it because we we prepared a lot for our session with you carl and and um kurt was asking some really open-ended questions that were that were on the on the surface quite naive questions but they elicited the most interesting responses from you so um it it really made us rethink a little bit how we how we do everything on our podcast yeah i suspect that part of that free ranging um beauty was so fast there was no hard limit of the time i had no idea three hours but he was he was a very engaging young man and i think um part of that open-endedness was deliberate you know he was he was just indulging a bit of epistemic foraging well it's funny you say that because that that's the free energy principle in a nutshell isn't it that we we need to balance the entropy back to this um this burden of knowledge because i want to i want to ask you a little bit about that which is um and i find this is often the case uh you know say for example um a kid comes and shows you a test that they're taking and there's a math question on there right now when i was their age i could have answered it quite easily but as i learned more about mathematics and engineering now i look at the question and i'm sitting there thinking yeah but there's all these assumptions like how do you know this variable's positive maybe it's negative like it could be a real number you know you have all this extra knowledge that's entering in and impinging if you will on the ability to answer that question and i think a lot of times when you're asked a deep question i i'm imagining because your depth of knowledge and breadth of knowledge is so large you're sitting here thinking okay how can i answer this in a way that's precise and and accurate and so therefore requires the use of technical and you know accurate and precise terms but then for the the person hearing that answer they may not have you know the the requisite knowledge of those terms and so it seems as if it's coming across as obscure or something like do you do you run into that do you run into that often yeah well i think we all do don't we i'm you know walking across disciplines um you've got to get that um you got to find the narrative that works you know in a sense if you're if you're taking a sort of um a free energy principle approach to this you're trying to establish some kind of generalized synchrony which you know so that my narrative becomes your narrative but that mutual understanding does require a commitment to the same rhetoric the same calculus right so you're often right a lot of theory of mind a lot of perspective taking um to the extent sometimes you have to sacrifice a degree of precision if i wanted to articulate something that i know i could simulate a write down mathematically it would just take um the kind of language which was not part of the portfolio the person i was talking to to go through properly so you know you just have to take shortcuts heuristic surprise and the like yeah which is part of the odds it is einstein keep everything as simple but as possible but not too simple that's the art of that well in a bit we're going to ask you about your most recent paper that literally has that title almost the uh the free energy principle made simpler but not too simple so we're going to ask you about that later indeed indeed we are and it's interesting what you say carl because you're talking about things which are so alien it might seem that you're pointing out the obvious when you talk about self-evidencing or when you talk about the basics of probability theory but keith and i often remark that a lot of people just don't understand probability theory a lot of very smart people and it it might seem that that we're reducing it down to the most basic um principles but they just can't be articulated uh very very clearly yes now yeah and people like me sort of lose sight of that you know there are some basics you assume people do know but of course it takes years before if you're fluent um and have all the right intuitions you know so that i think in part is um an explanation for why people like me are so often cast as being impenetrable or very difficult to understand when it's not but you know it probably reflects again a lack of theory of mind in terms of being able to you know understand what people do know and don't know wonderful well i'm going to officially introduce you carl professor fristen is a british neuroscientist at university college london and an authority on brain imaging in 2016 he was ranked the most influential neuroscientist on semantic scholar he pioneered and developed the single most powerful technique for analyzing the results of brain imaging studies and he's currently a welcome trust principal fellow and scientific director of the welcome trust center for neuroimaging his main contribution to theoretical neurobiology is the free energy principle and its bayesian reinforcement learning flavor active inference the free energy principle is a formal statement that the existential imperative of any system which survives in the changing world can be cast as an inference problem the probability of existing as the evidence that you exist if you will this is what carl refers to as self-evidencing or even autopiosis now the system of course could exist at any scale or time and as you might well imagine this will lead us directly into a discussion on complex systems theory self-organization and emergence anyway and professor fristen it's an absolute honor to have you back on the show welcome thank you very much it's great to be back talking to you again oh and incidentally a great summary of the free energy principle was that from wikipedia or something slightly more learned well we we like to interpolate from multiple sources yeah um you know if we can just jump back actually to uh what we were what we were talking about with um you know things that are simple that are being that may still be hard to understand um i i if i may i want to i want to share an anecdote with you and and this is really this came to mind when i was reading a recent paper the free energy principle made simpler but but not too simple and the first line of the introduction says it is said that the free energy principle is difficult to understand this is ironic on three counts first the free energy principle is so simple that it is almost tautological and what occurred to me there was one of the most valuable life lessons i got in graduate school was i was in a cryptography and cryptanalysis class and it was being taught by professor sylvio mccauley and he's since gone on to quite some fame in the in the crypto cryptographic space and the first day of the course okay he puts a secure communication problem on the board the very first thing that he does on that day puts a secure communication these are kind of like brain teasers in a way you have to come up with a way to solve this problem and he goes okay does anybody know how to do this somebody raises their their hand and gives them an idea and he goes no that won't work here's why and he shows how to crack that a couple minutes later another student how about this uh two random numbers communicate no that won't work either here's how to crack that right and so after about uh five or six attempts from the class maybe 15 or 20 minutes nobody could solve it and he goes okay this is the perfect place to begin this course because i'm going to show you the answer and when i do it's going to be so simple that you're going to say ha that's easy no something that's easy is easy from the beginning if somebody has to tell you the answer it's only simple right and i think i think you know human human history is is just full of examples of ideas that are very very simple yet took you know the combined intelligence of humanity thousands of years to discover they may be very simple ideas but they're still very difficult to kind of grok to really internalize and and understand i think we have those all over science and the free energy principle may well be one of those so even though it may be very simple it could be difficult to kind of cognitively grasp but i almost i'm almost starting to think over the last year or so that human cognition itself like in the cognitive space there's there's sort of this markov boundary between an external world of ideas and and internal and there's a fluctuation there at the boundary where sometimes the most advanced ideas they may creep into what's in our sensory range and we can just grasp them but then they kind of fly out again and they're just always on that cusp of what we can really understand just curious what you what you think about that i i think that's a very compelling formulation of the and i've never heard that before the the distinction between easy and simple as you were talking um i was mindful that everything you said could in in one way be applied to evolution um uh and indeed in terms of the yo the um evolution of ideas and the scientific prog uh process itself cast as in cultured evolution you know sort of co-evolution or evolutionary psychology um as one special case of the evolutionary process you could argue that um evolution is a really solving a really difficult problem to disclose a really simple solution and that simple solution of course is you and me just finding the simplest phenotype artifact that fits with the eco niche that is co-constructed by other artifacts and simple uh um constructs markov blankets um that constitute that very condition i think that's a wonderful insight um and the simplicity bit i think is really quite crucial um in the sense that um well let's take the uh the cryptography or crypto analysis um i'm mindful of one of the very first introductions of variational free energy um by david mckay who just applied free energy minimization to the problem of encoding and critical analysis from the perspective of minimum description lengths and compression uh which speaks to you know an alternative view of the imperatives for any efficient representation or evidence accumulation or data dissemination scheme or communication scheme um in the sense i mean that in the sense that you can either read the history of this from richard fireman and the pathetical formulation of onto electrodynamics or you can go right back to algorithmic complexity comodo graph uh complexity sort of induction through to minimum description and compressibility right through to universal computation both at their heart have this notion of simplification as aspiring to or converging to a minimum complexity a minimum description and a maximum compressibility so if you read you and me as the products of evolution that are the most efficient installation of all the information that i need to engage with this eco niche that we have co-constructed that exactly the same principle emerges i guess what you're saying is and i think you're absolutely right that they those kinds of principles not only applied to our structural phenotypes and the functional architectures that we have in our brain and indeed uh our bodies but also the cultured constructs what people like uh cecilia hayes would call cognitive gadgets you know the things that um beyond a canoe which has evolved but in a way very differently from the way that you and i have involved uh beyond that to things like language itself you know where did language come from how does that simplify our sense making an efficient representation of causal structures structure in the world so i think there's some really profound common themes cross-cutting themes in that description and in a sense of course spotting which is not easy but spotting those simple cross-cutting themes is exactly the process that would they are that you're talking about right right and i and i often get the feeling that um many of us or or many researchers out there are all orbiting this kind of central truth and it's not just a there are the cultural constructs the physical constructs and some abstract concepts that are somehow grounded in the core you know reality of physics and in the universe and i feel like we're all orbiting them and we haven't quite gotten there yet i think the free energy principle may be the closest you know but there's still there's still gaps there's still things that we don't understand they're still missing parts and it'll be really interesting when when somebody finally locks them all together and grasps that that simple connection of them all so how will you know when when they've done that uh that's a good question how would they how would you know when you've done that i guess for me it would be when we really have a mathematics uh when we really have a mathematics of cognition so it's it's close but but still far away in a sense yeah i agree entirely yeah and that simple point having that um i mean if you're committed to this notion that there is um an evolutionary process in play um which in my world i would read as nature's way or natural bayesian selection or optimizing your explanations for the world with respect to the evidence the basic model evidence format so i look at natural selection as you know the uh the process of uh basic model selection and interestingly just coming back to your example uh in your first class with with york and well-respected professor he was illustrating their i think the process of basic model selection by putting forward alternative hypotheses eliciting them from the audience and then evaluating the evidence for them by in this instance falsely finally because that particular strategy was easy to crack but illustrating that this is a process of selection and evincing a deep problem which is generating the hypotheses the models the phenotypes from which to select so i think that that you know that that's why i think the evolutionary us evolution perspective is it's so compelling you know speaking to issues that the difficult problems which i think in radical constructivism would be known as structure learning you know it's easy to score a good structure a good idea a good explanation a good hypothesis a good bit of um code algorithm it's easy to score how good it is but to actually create a portfolio or an ensemble or of alternative to build a hypothesis space in the first place that's the uneasy but that's a difficult bit because what we're talking about is just being a good scientist it's asking the right questions by having the right kinds of hypotheses for which and search search search work for evidence that again that was a very nice example to um yeah you know to bring to the table now completely lost what we were talking about what what did you bring to the table with your question oh no no it was just it was just a comment that um that we all should should take care that simple concepts are are can be very very hard and furthermore simple concepts can be extremely valuable so the the the the two lessons i learned there uh were that simple and easy are not synonyms and simple and trivial are not synonyms and you also um highlighted the importance of a precise and simple way of expressing those ideas in terms of mathematics but i think and what i'll i'm tempted to ask you but i i i won't um you know but why mathematics um and mathematics is the is the simplest most self-consistent calculus language that you you could adopt furthermore it is the kind of language that you could um use to communicate and share in the sense of this um cultural approach to co-evolution uh with people who don't speak your native language with your uh vietnamese colleagues or you know people who don't actually communicate with narrative form but you can still in a very precise unambiguous simple way express those ideas and those hypotheses even mathematically and it will still be understood at how meaningful somebody um in another culture in another world and you know speaking speaking another language so i on that view if you um had to commit to a particular language that it really has to be mathematics which is actually a bit worrying in the sense not everybody is mathematically fluent um which brings us back to you you have to understand certain simple mathematical fundamentals before you can even start talking yeah although interestingly it's a little bit like tetris so you don't you don't need to understand that many fundaments about maths until it has an extrapolative effect in many different parts of experience space if you like but um professor fristen i wanted to talk a little bit about the the low road and the high road um you often say that the low road is the mechanistic view of predictive coding and neurons in the brain and all the low-level stuff the high-level road being the emergent and the self-organizing phenomena um let's just do the low road first we want to get to the high road later but um we just had a wonderful conversation with professor joshua benzio the godfather of deep learning and he's doing something very similar to you um he's got this um this invention called g flow nets and he's using that in place of the variational inference technique that you're using in active inference to compute the free energy g flow nets can be thought of as a drop-in replacement for markov chain monte carlo for computing unnormalized energy functions but without the hand crafted priors and um even if the modes and the energy function are thin and far between and in a high dimensional setting it's a kind of machine learned version i think you'd find it very interesting but anyway um benjio is also a huge fan of this distributional reinforcement learning and balancing entropy in a principled way and when i pointed out the obvious similarities to your work and active inference he spoke very positively so you need to check that episode out lovely i will do so yeah i've never heard of that but that yo i have heard him talk before i do get a sense he's you know he is certainly becoming more of a physicist as he gets older the probabilistic physicist i should say yeah that's that's very nice so what's changed in the last year for the free energy principle let me think well one thing that um interestingly we've been working on is indeed this communication and trying to sort of um get towards a simpler exposition that people can already readily access and use in their own in their own setting um part of this is uh until spotting what the important simple contributions are and what it can do what it can't do what it's there for what it's not there for um trying to pair back the found the fundamental parts of it from those um secondary auxiliary issues which are more interpretational in nature so from a technical point of view um it's just been staring at the ideas thinking about the ideas and seeing what is the absolute essence of this and trying to articulate that mathematically as simply as possible but um not making it too the the one technical aspect here um is what is the precise role of the markov blanket um and to what extent does that rest upon the notions that you would inherit from complex um dynamical systems theory relative to a more probabilistic uh construction of the kind that say and judez pearl's uh formulation would speak to um probably more importantly how does one relate to the other so how does a sparse coupling of a dynamical sword lead to certain forms of conditional independence those conditional independences then underwrite the um the probabilistic formulation of the coupling between inside and outside of a of blanket which is at the heart of the high road to and to thickness and self-evidencing and the free energy principle so that's certainly been a focus um i have to say slightly inspired by skepticism in not from mathematicians or machine learning or um artificial intelligence but from philosophy so philosophers become very exercised about these things um and like to drill down on the um the precise implications of um anything that they understand when they when they read the map and certainly one thing which has caused puzzlement in that field how do you constitute or where do you place a mathematical description in terms of complex the dynamical systems theory and in particular the theory of random dynamical systems how do you place that in relation to more conventional conditional independence uh sort of perlion um uh markov blanket so a lot of work has been has been focused on that and just for your interest they have the upshot of that is possibly going to be a move away from casting things in terms of uh non-equilibrium steady states casting things in terms of the probability distribution of a system a markov blanketed system at t equals infinity as the underlying if you like aspiration the probability densities that prescribe and shape behavior um a move back to the much simpler formulation in terms of lagrangians and path integrals so this this actually is a nice reflection of this trying to get back to the to the the kernel and the simplicity and interestingly in this instance uh right back to the inception of um certainly on my reading of history uh of variation free energy in the context of the package formulation so just getting right back to what is the lagrangian what is the generative model under the hood how is that manifest in the equations of motion and what does that look like in terms of the implicit information geometries that come from the probabilistic concomitants of the these random dynamical systems so i would i imagine you you were talking previously about um the most recent attempt to simplify demystified free energy principle um uh before that paper was actually put on uh archives there's another one which is even simpler much much simpler that just excuse um all the delicate issues of non-equilibrium or non-equilibrium steady state intensities and just sticks and tries to pursue the whole story from uh front to up to end using purely the notion of a lagrangian and pathological form of formulations so technically that probably isn't terribly interesting from your point of view or possibly even your viewers but but mathematically it's it it's been a real game getting back again that difficult process of getting it simpler and simpler and simpler and so that's that's only one thing that's changed if you're asking me what's happened um in terms of uptake and um application of these ideas um then i think yeah i think things are moving more quickly than they were um a year ago and i'm talking here about the application of or the corollary in terms of your process theories of the way things work or the way that you might want to build artifacts or algorithms um which is the active inference so certainly there's been a much greater uptake and interest in active inference um that i've seen simply by pressuring number of emails and and conferences and workshops so literally um i think two or three days ago there was um a wonderful uh workshop at the berlin school of mind and brain which was just about active imprints and free energy principle and a bit of philosophy at the end which was you know interesting mixture a cocktail of intellectual challenges but you know i was really impressed by work which i had no idea about that was you know ranging from um simulations of declarative memory actually getting little uh synthetic subjects in a computer to actually told you why they made a choice so you're taking sort of a fairly standard um economic games or the kind of paradigms that people in reinforcement that might their learning might use but having an epistemic aspect so seeking for information to optimize their decisions and their choices later on but then taking um the the active inference framework and then putting on top of them the ability to debare and explain why these decisions were made leveraging you know what distinguishes active inference from reinforcement learning which is that under the hood you have an explicit representation of beliefs belief structures so there is a belief based approach to any uh any decision or any move that you might make and you can use that to actually get certain choices of decisions and actions to disclose the belief state so you essentially you can give again the computer to tell you why why i did that so that was a wonderful example great stuff going on in in robotics and your robotics um well i say great because of myself colloquial uh probably sort of quite narrow view of it but i'm really impressed with you know what these people are doing and what they are doing is just taking as rat off the shelf the um the particular objective function that underwrites active inference which is the expected free energy and then thinking well okay how can we scale this up how can we make this work and they're bringing things like amortization and all the latest um machinery from learning machine learning to amortise some of the more delicate problems of mapping say from high dimensional inputs uh to the beliefs as quantified by the sufficient statistics of uh of distributions very much in the spirit of a variational autoencoder that you know that actually has um at some level a very simple belief structure or um posterior belief um and associated with the you know with the equipment the values of certain nodes in certain layers but taking it into the realm of state space models where you know you've actually got an ongoing dynamics and engagement as the rover has to have with the environment and of course the added and beautifully challenging um aspect of robotics which is the inactive aspect that it's not just a question of doing uh making sense of the data quickly and efficiently in a belief-based way but using that kind of data simulation in order to decide what's the best next move that will enable you to assimilate your data better under certain constraints which of course you know um are particularly present in your in robotics so that that was very interesting um i've noticed um there's been a move coming back to you know to the original um the original theme of of communication and uh mutual understanding between disciplines and demand between people trying to take these ideas forward and converge on the the endpoint which is which we might might not know where we've got but this simple practical things like much of in my world much of the um the proof of principle and the the demos that tried to unpack and add render transparent and accessible the basic ideas are written in academic uh software usually matlab and i wasn't aware of this but no one else uses matlab anymore i'm free millennial but apparently you meant to use python and julia and all sorts of things i'm afraid well actually you you know hey use whatever tool whatever tool works um well first of all i i'm excited to see that that you're seeing more uptake of the free energy principle because i think you know both tim and i after after our show last year have you now now we're seeing it everywhere like i mean we're seeing the concept everywhere it's like everywhere we look we think oh this is there's a connection here between what we're talking about now in the free energy principles so i'm certainly a fan of people definitely thinking about it more trying to come up with uh practical systems based off of based off of the principle um i want to come back a bit to what you first talked about which is the mathematical developments or exposition if you will of the free energy principle um and i did mention you know the the free energy principle made simpler but not too simpler which is apparently yet a less simple version than a simpler version that you had also published which i haven't looked at but what i loved about the paper that i did did look over is the formulation of um of this as stochastic dynamical systems that are you know orbiting around a pullback attractor you know so so so they are trying to get to this this attractor of course they're never going to arrive there and that was a very important point that you made in the paper that that hey look uh you know this idea that uh let's say life is conducting and optimization is a bit technically incorrect you know it's not doing a formal uh optimization over the entire marginal distribution it has a set of dynamics that it follows so there's a dynamical system that's kind of going let's say in a sense downhill you know towards this this um optimum but it's moving around it's it's being uh perturbed by external variation etc so you get this kind of wandering around around the attractor and you said that you know philosophers get very worried about how do you connect um you know systems like that to to inference or to probable inference and you give a connection i believe at least if i read correctly in that paper you know that you did actually give a mathematical connection between let's let's show how you can take the dynamics of the system and map it exactly to inference so i believe that connection is provided in there is that correct no that's absolutely right and indeed it is just that connection that is the free energy principle that you're that and in a sense um well in the sense that we're talking about um it does not surprise me that you are seeing that very simple construct emerge in many many different fields and many guises because that's the whole point of just um trying to evince something that has this cross-cutting aspect that you should for anything that works in the sense that it persists um uh in the context of which you place it then it must possess these properties so what what you've just said which was a very concise and erudite summary of the contribution of that paper is just an expression of things that possess persist or things that exist so you know how would you write that down mathematically well if they exist over a certain period of time then they have to have an attracting set what kind of attracting set are we interested in well we want the most generic um accommodating mathematical description of systems um that does not um actually just qualify this that does not go quite as far as a quantum mechanic background-free uh description but as close as you can get without going without going you know crossing the going to the other side um and that's a random dynamical system so what is the attracting set in a random directional system it's a pullback attractor so you're starting off with well okay so we're talking about things that exist in any given context in any given setting things thickness has to be defined that's the markov blanket what does um what does that imply having a mark of blanket well if it exists sufficiently long for you to be able to and uh for the conditional dependencies to be expressed so that you can write down there or assert that there is a a markov blanket then you have to have a degree of persistence or existence so thickness entails persistence over time that entails and necessitates them the pullback attractor just chill picking up on your uh on the sort of um the the possibility of um over interpreting the role of um of a pullback attractor in the sense that of course you know the definition of a pullback attractor would really be the attracting set that um attracts all states uh as t goes to infinity is that a really relevant construct when we're talking about real systems that have the an itinerary a separation of temple scales um and as soon as you invoke a separation of temporal scales which you would have to to talk about the difference between inference and learning learning and development development and evolution um are you licensed now to restrict yourself to a random dynamical system of the kind that could be written down by a launchman system with an attracting set that may that requires equals infinity for its um for its for its definition and clearly not you have to actually sort of confront certain interpretations of how you are using this notion of a pullback attractor um and that's certainly been something that's been going on in the past year or so now lodging conversation i repeat with people who are mathematically fluent but their primary agenda is more an interpretation and philosophical interpretation uh and what's come out of that uh or um certainly it's one one um motivation for that kind of discussion uh is this notion of separation of temples time scales so in a nutshell um the uh we've stopped saying talking about self-organization to non-equilibrium steady state because that implies that we're waiting until a system has reached t equals infinity and is at steady state and that's never what was um in mind we were using the steady state and the notion of an attractive set simply to say that this set of differential equations um has a solution that is non-trivial in the sense that it does the states don't go off to personalize so if they don't go off to plasmas infinity as time proceeds then there must be an attracting center must be a solution a non-trivial solution and it so happens what could interpret that non-trivial solution in terms of a non-equilibrium steady state solution if you've got a markov blanket that endows it with um the fingers and you've got solenoidal flow uh to make it a non-equilibrium as opposed to any caribbean uh steady state and then the the question comes um okay so if you're just using the notion of a pullback attractor instead and it's implicit uh step state solution to the dead city dynamics does do the density dynamics of themselves um have any um mathematical or theological role in them and can they possibly if they only if this the object is only defined in the infinite future so what you do that says well no no all we meant was for a suitable period of time there are certain equations of motion certain dynamics at play that have a solution and that the the the way that you get to a probabilistic understanding um of the behaviors and the dynamics that this set of equations or these and this dynamics has is just by um uh looking at their their underlying lagrangian uh and looking at the uh the dependencies not um in terms of the states occupied but in uh purely from the points of view of the paths that the system takes for this period of time and then that affords the opportunity i think this i should have mentioned this when when last year what's what's been happening recently uh from point of view the people the free energy principled then you say well what do you mean by a certain period of time well then you get now to the separation temple scales so you have this notion now that you can understand the the paths the trajectories the dynamics um of this thing that is defined by markov packet for this period of time and then you say well let's make the equations of motion drift at a slower tempo scale and then let's repeat the game and how will we do that formally well we'd use the renormalization group or the apparatus of the renormalization group and then what you have is now you've now freed yourself from the markovic constraint so that now the universe has now articulated in terms of these series of random differential equations where the random differential equation at any one level supplies the control parameters of the parameters of the equations of motion for the dynamics at the level below for a period of time means that you've now have the opportunity to look at how the realization of a free energy principle say in terms of active inference uh at one level is deeply literally contextualized in a hierarchical or renormalized renormalization sense by exactly the same form and dynamics at the level above so now we've come back to evolution now we've got a slow process in fact that's a little bit grand now we've got let's make it slightly simpler now we've got a nice um mathematical distinction between inference and learning uh and i'm focused on that because clearly many of your your viewers will be watching the machine learning and advances and in that field um and what this brings to the table now this uh implicit separation temporal scales is quite a fundamental difference between interfering and learning immediately tells you that to optimize the weights of a deep neural network or the parameters of variational auto encoder or germantown the serial network you have to do it after you've done some data simulation after you've done some so because you've got the separation of temporal scale both at each scale exactly the same principles apply and can be both formulated in terms of self-evidencing just maximizing the marginal likelihood at that level but the connection between the two now enables you to think about how learning can contextualize inference and how inference supports learning so um by inference here i simply mean um optimizing beliefs or maximizing the marginal likelihood of bayesian modern nebulas or its elbow or its variation free energy pounds um with respect to belief distributions or probability disputions over things that change quickly states of the world so the kinds of things that you would do you would need to infer when doing surveillance or doing um your speech understanding or update in an active influence sense uh speech production um as opposed to those things that fluctuate slowly which are the connection strengths the weights the parameters the laws the contingencies that are in play in this particular context noting this you know there's a could be another level above which is a slow change church of the contest that you can learn and then you can start to understand right from the helmholtz machine right through to amortization your notions of learning to infer this is optimizing the slow process to make the fast process optimal um um and uh and as such noting that the the way in which you optimize the fast the slow stuff depends very much on having uh some optimal um optimal inference scheme or digital simulation scheme at the past level pursuing that argument it also tells you well the stuff all the way down and stuff all the way up in terms of separation table scale so you naturally now start to ask well what's the scale above the learning and this would bring us back to this evolution and model selection structure learning yeah how many nodes how many factors you know what's the actual structure do i use a convolutional neural network yeah all of these are structural considerations that you could use for example a genetic algorithm or a launch of an assisted mcmc at a very very slow type scale to now optimize the very structure of your variational autoencoder that's doing the learning to infer that's enabling inference in a very fast time scale that leads to an interesting idea so uh you know having different time scales different levels of of the structural learning and you know i don't know if that's been tried out yet but i think it's it's definitely worth worth trying um i i think i i have a different question though about about the paper and thank you for that exposition i was very helpful one thing that's kind of occurring to me now is that again the concept of the markov blankets are still you know fundamental and one like perhaps concern i don't know if i should say concern that i had with the paper is the way in which it it formulated the uh markov blankets you know in terms of the the curvature of the surprise being zero like at least you know you you uh as people may remember um you know the sensory sets are those sets which the external world um can can uh can control and you can view but you don't control and your uh your active states are the ones that you can control not the external world and the external world can view them and so these there are these sort of four regions and in the paper you define you know ways in which let's say you can set various terms to zero which mark the you know the boundaries between those those regions what i'm kind of wondering is do they have to be zero like what if um what if this is generalized a bit to have like a degree of active stateness and a degree of sensory stateness and there was kind of a more smooth boundary between these what would happen i mean would that give us any useful useful outcomes mathematically or would it just get in the way and it's just more convenient to continue you know zeroing out certain elements of the matrix like has any has any thought been put into what if it was more of a matter of degree and would that be helpful or just get in the way now that's a very astute question and yeah i think when a lot of thought has been um directed at that problem whether there have been any useful answers and that's that that is another another question um so just if i can um paraphrase your questions i think a really important one what you are saying um is that before we make any moves in terms of understanding the dynamics of things that persist in any given context we have to define things and in this particular construction we are defining thingness in terms of sparse influences um that um underwrite conditional independences so the your the zero elements that you were talking about there are absolutely crucial um in um defining the thing um in a crisp and unambiguous way via conditional independence is that just technically for those people who like thinking that in these terms and zero entries in in the hessian of uh of the system when when um where the hessian is the um the curvature or the um the second order uh derivative of the of the of the of the likelihood effectively um and then the big question comes then where did these zero entries come in that defined stipulatively thing that's in terms of conditional dependence if it was the case that um you could you um could not um establish conditional independence between the inside of something and the outside of something then there is no way of really separating the states that into inside and outside the game then um and the delicate game and and the one which has been receiving attention the last in the last few years in the purely academic philosophical literature but also the maths literature is exactly this question of how does sparse dynamical coupling mainly i can't actually plus you because you two are too far away or you don't the right kinds of receptors how does that translate into these zero entries and then your question is well okay do we have to be scam commit to exactly zero can we actually now have a little bit of um wandering away from these hard constraints um now mathematically i wouldn't want to do that because as you say is much more convenient just to sort of uh identify um zero entries in the hessian and implicit conditional dependencies um under some sparse coupling so what i would say i got two levels the answer first of all the out the the zero the absence of any conditional uh or the presence of a conditional and is not a terribly restrictive or magical um requirement in the sense that it is just a reflection of sparse coupling so i think that you would put the question really not to the dependencies and the hessian but really to the notion of um an influence diagram that was predicated on a set of differential equations you thought was appropriate to describe this universe now if you think universes have all to all influences so every state of this universe come in a dynamical sense simply the sense of a launch of an equation or a random dynamical system influence every other system then you will have a suit and you will have you will have that may have interesting structure with it but that structure will not pertain to anything that can be separated from the universe because everything's connected to everything else so you start asking well well can i just can i jump in here because i think um to a degree much of the universe and just kind of leaving aside okay i'm sure there are some you know horizons at which locality you know locality horizons whatever but but i think that on the scales of say human beings we are in a soup i mean we are influenced it's just those influences may be very very small you know almost almost epsilon in a sense um and that's what kind of worried me about it is that look i i think i think i know that at least in terms of physics you know there is this this soupy influencing of everything kind of on everything else and so what happens if we're dealing instead with mathematics that has these sharp you know zero boundaries is it going to be an act an accurate reflection of the more soupy albeit still exponentially decaying you know uh couplings perhaps i should just ask you i mean certainly it is the case with um six degrees of separation that there is a vicarious influence that that you know will almost certainly be non-zero but though did you mean those vicarious influences because for me of course the the whole notion of a vicarious influence through something else immediately invokes the notion of some thick which means another mark of blanket so certainly there can be indirect influences and i'm not denying that i'm just in terms of the physical um it's probably a bad word he is but just in terms of the functional form of the differential equations that will evidence this precarious all-to-all coupling by other states do you think that um that most universes uh are sparse or not sparse um so with the assumption is here that it is the sparsity that pro that endows this rather direct system with structure and in fact it the exception is a non-sparse connection um and then the question is how do you get soupy mutual influence amongst all the constituents or states that that constitute that randomical system so you know if you didn't have that kind of sparser you would never have a hierarchical organization in the sense that a hierarchy just is the lack of an influence transcends more than one level say the subsumption uh hierarchy um in physics and well let me let me before i forget that the other half of your argument is much more accommodating of the epsilon and and the fuzziness which which brings us to some notion of wandering sets and back to um and back to the uh this notion of separation temple scale so how long is it nearly zero um or almost zero and acknowledging that at some point it's going to be not zero again because things change so that i've jumped to the second half of the argument but i'm interested to try and bring the the really important uh from my perspective notion of sparsity back into focus and say that it is not um a remarkable um to um to assume lots of zeros in influence because we do that all the time when we write down any differential equations just by emitting one variable um in a state vector from the function that defines the equation the motion the flow operator as soon as you omit one of a universe of um of variables you've introduced a zero um a zero um um in that uh in a sparsely in that dynamical coupling maybe you know maybe this is really the heart of abstraction which is even if there are these very tiny soupy couplings we can abstract we can just treat those all zero eliminate them and still that abstracted model um reflects you know emergent behaviors of of the things if you will of the universe really that we're trying to model and i think that's kind of this duality between between the fact that we may have a soupy substrate and yet still discrete uh sparse behaviors emerge on top of that that soupy substrate is probably one of the greatest mysteries for me so i don't know the answer but i i just wanted to get your your thoughts on it you know about well i mean i think that's a very important um the question um um i i've seen it asked um interestingly enough um um in corresponds with people who have sort of um trying to look at the origins of life and i tried to understand the emergence of of biotic self-organization and from many different perspectives you know from uh the kind of approach that people like kate jeffries or jeremy england might might take right through to um uh theoreticians think uh to try to reproduce um organic like self-organization by reproducing you know the events of the bottom of the sea you know april uh primordial soups in structured containers afforded by these vents one thing which struck me um in these arguments was the notion of emergent behavior that had this self-replicating aspect to it from a dynamical systems perspective the emergence of attractors that um have a poincare section you get sort of um a cycle of life emerging was the notion of shielding the notion of protecting yourself from the influence of other things um you know um as a way of understanding uh enzymes for example or ways of understanding the physical structures that lead to sort of uh human sort of cell light formation formations that notion of shielding comes back to the possibility that it is emergence of sparsity and the construction and the self-construction and the autopoiesis of sparsity the underwrites a markov blanket building system that is necessary for the kind of structured self-organization that um that you're asking the question about so literally how do we move from a primordial soup a set of discernible things and the answer seems to be from what i'd read it's the emergence of certain and um auto as uh auto catalytic uh structures to actually sequester and shield certain sort of chemicals or processes from other chemicals and in the shielding thereby emerge the supply point of view the mark of blankets so again we come back to sculpting structure by removing stuff in this instance it's removing connections creating the right kind of sparsity where structure actually emerges from the sun point of interest um cole could you define i'm really interested in in your definition of self-organization and emergence and and also do you think it's a useful um distinction to have this notion of strong emergence and weak emergence could you just give us your take on i could if i was more of a scholar but i'm going to have to ask you to define stronghold with emergence well it's all about reduce abilities i mean i think weak emergence is a kind of emergence where the emergent property is amenable to computer simulation or you know similar forms of after the fact analysis so like you know the formation of a traffic jam and strong emergence has this notion that the whole is something more than the sum of its parts i mean i i was actually going to ask you a slightly related question about substance dualism right which is that you know some philosophers have this notion of um the mind not being reducible to its component parts so a materialist for example or a physicalist would would think that do you think there's a kind of parallel with this notion of strong emergence and the irreducibility of our mind and consciousness to the uh constituent components right you you're clearly more versed in the in these uh fields than i am so i'm not going to pretend to give you i highly doubt it can't get this instance you did a little more than i did however i get a sense if if you wanted to proper answer that question i think it would probably come back to this um this separation temple scales um and posing that question in there to the kinds of systems that would uh that would constitute a renormalization group and say is there anything that um emerges at one level that could be abstracted from the level below um if you could find an answer um that satisfies you in terms of the difference between strong and weak emergence by looking at a system that has that renormalization property with the separation temporal scales then i can i could tell you what the answer is unfortunately i don't know because i'm not sure what the commitments of strong and weak emergents are but certainly it would be impossible to understand um one temporal scale or one level uh without um um writing down art simulator understanding um the level above and the level below so in that sense i said i suspect that it might there might be a strong emergence in play here the reason i mentioned that is that of course is is um is the get out and it's the mechanics it would appeal to to resolve the um the the sparsity or the hessian equals zero problem so it's clearly the case that uh even in a soup or a soup that transit or a collection of well-defined cells where where there is this shielding and um uh sparsity of influence that protects or insulates the internal dynamics of say a cell from the external milia at some point there's going to be cell division at some point there could be some uh metamorphosis or transformation or phase transition where it is rendered soupy again and you would need a different set of markov blankets a how do you go from one cell to two cells so you've got this notion now of wandering sense so the markov blanket partition was nicely described into a as a partition into internal and external and the intervening active and sensory states that mediate the bi-directional influence of the inside the outside that partition comprises a subset so what you're talking now is about the possibility of wandering sets in in the sense of burkhoff how on earth can a set wander when it is defined stipulatively in terms of sparse coupling expressed as a random differential equation or a longevity equation well it can change if you say well this um set of equations of motion is only in play for this amount of time and then the next chunk we're going to change the parameters of the equation of the motion and bring it other states as soon as you bring it other states into your flow operators at a slow time scale you're now destroying the old sparsity structure replacing it with a new sparsity structure or possibly destroying this sparsity structure altogether so i think that that would be the way that you put putting excilon back into the game but actually more than epsilon actually allowing for the fact that there is a certain scale of organization in space and time uh that is superordinate to the you know the microscopic dynamics of your your level of inquiry over which you will now have a reconfiguration of the markov blanket so things will look as if they're reorganizing uh so what level of self-organization happening not that you can't answer that question it's at all levels because the the context of self-organization at one spatial temporal scale depends sensitively and critically on the parameters that define the initial conditions from the the flow that are inherited from scale above and likewise the scale above has to conform to exactly the same dynamics it has to have things in it so i think that's how you get out of the getting to a more plastic um semi-markovian universe a biological universe you'd start to answer the real questions of orthopedists i i feel a little bit um naive in saying that the free energy principle does autophases um in retrospect because order points has a lot more to it and it's exactly what you've been talking about it you know how do you kept these self-preserving how does how do the fast microscopic dynamics create the context for there to be a self-assembled mark of blanket that entails the fast the fast the faster these are really deep questions indeed it it it's another example actually we um we've spoken to cyberneticists like um professor jay mark bishop and and um he he introduced us to lots of stuff about otto poisson um i've just got a bit of an eye on the time so we've got a couple more questions to get through but that was absolutely fascinating professor preston but um i'm not sure if you're familiar with ilya suzuki he's the chief scientist at openai and he became embroiled in a heated twitter debate a couple of weeks ago where he asserted that large language models you know the likes of gpt3 might be slightly conscious and of course i'm speaking to an authority on consciousness here when when you spoke about consciousness actually recently you said and i'm quoting um the proposal on offer here is that the mind comes into being when self-evidencing has a temporal thickness or counter-factual depth which grounds inferences about the consequences of my action in this view consciousness is nothing more than inference about my future namely the self-evidencing consequences of what i do i think this is a fascinating way to put it and keith told me earlier that he thinks of consciousness in a very similar way but the one thing that is missing from this discussion in my view and by the way we're speaking with um good old david chalmers in a few weeks time but what about the phenomenological characteristics of consciousness uh right so we've got a couple of minutes to do that that that'll be fine we could get david to watch this uh give him my very best wishes oh we certainly will and and by the way if you have any questions as well for us to ask david on your behalf i think that would be very well you should ask if everything you've just asked me and see what he says i love baby and i also like the way he responds to some of the more technical questions that he gets and his response to this i think um would probably um loop in the notion of the meta problem or the meta hard problem um which puts simply in the words of andy clark would be why is it that certain things i.e philosophers and people like you and me puzzle so much about our qualitative experience and and it may sound a strange uh question but it's very revealing question it simply means that if you subscribe to the notion that we are inference machines we are built to actively self evidence then you are saying that we entail um a generative model of our experienced and if we have a dirty model of our experienced world and we can uh talk about and puzzle about cult of experience we must also have a hypothesis that first of all we are things second we are having a court of experience with the alternate that we are not um which of course then um provides a really simple explanation as to why philosophers uh love things like brain and vats and the zombie arguments yeah irrespective of whether they are useful thoughts the very fact that you can get some like david chavez as a young man talking about zombies and people listening tells you immediately you've come across a certain sentient artifact on on earth that has the uh has got a generative model that can entertain the counterfactual hypothesis that they don't have quantitative experience now you may be asking well why is that so interesting well it's interesting because it tells you immediately that these kinds of general these kinds of centered creatures are showing up um ascension behavior necessarily requires the ability to hold counterfactuals in mind and i mean so now i'm trying to work back to your introduction of the notion of the temporal depth and something that anil seth might also complement with a counter factual richness so the argument here is very very simple and which i'm sure you you know but they're probably worth rehearsing um so if you had a um a program um that could understand language then what would be the requirements of the underlying generative moral that it was using to infer the um the to make sense of the auditory stream to infer that the meant word well for it to be conscious it would actually have to be able to ask questions so we'd also have to be able to speak um so you know that's the the first inactive move that you know you you see with the triple e since the the turn of the century so we're talking about systems that are not just sentient in a passive sessile wave but have sentient behavior so there has to be behavior so your language understanding machine has to be able to do something it has to be able to move or ask a question and of course then you ask them what are the best questions well those are the ones that comply with the principles a bit up to a bayesian design and then you can trace out right back to things like or from my point of view uh articulate that as a special uh warm part of expected free energy but the key thing here is that the um the the the program of the scheme the algorithm has to be able to move ask a question interrogate the world uh uh perform an experiment on the world to do self evidencing if it's doing that then it has to have a generation model of the consequences of its moves so that's the simple and big move here that just if you want to talk about um sentient behavior you are necessarily if you subscribe to the fact it's all it all entails the generative model then you're talking about things that have genitive models of the consequences of um their moves their behaviors their choices those uh consequences are in the future they um they are necessarily um acquire that the genetic model necessarily acquires a temporal depth so just saying that i have an understanding or internal model with your psychology could be a simulation model um and if you're a motor controller it could be a you know sort of a forward model um but you're most generally just having a genetic model of the consequences of action means you've got a temporary deep genitive model and notice it's the consequences of your actions so there's a there's an implicit agency that suddenly comes in into focus so now you can use the word agent as soon as your genitive model contains the consequences of action um then you can use the word agent you can't do that can i just pick up on this a little bit it sounds a lot like you're saying um it must be a closed loop process and this rather leads back to i know you were talking about um things like aixi and and some of the the information theoretical conceptions of intelligence and and those conceptions also rely on this notion of of an agent acting in an environment does does that seem in any way i don't know it it feels a bit um anthropocentric doesn't it to use that notion of consciousness or or even intelligence is that a good framing yeah i i well i hadn't quite got to consciousness at this stage uh which which will um we will be certainly now a and in a way that um if you like celebrates that anthropomorphic aspect all i said at the moment is that if you want to talk about um sentient behavior um before you go to consciousness you've got to have temporal depth you and crucially um in having consequences there can be more than one outcome to any act so now you necessarily have a counterfactual aspect to the things in your journey immoral which brings us to my point before about you know perhaps the best thing that um david charmans presents from my point of view as an existence proof of a sentient creature that can have counterfactuals of a certain kind where it's not being able to have qualitative experience but this is an example of a counterfactual hypothesis this is a gift of very very sophisticated uh systems that are probably a long way down the path to ultimate simplicity uh in terms of evolution uh or to a very very slow type scale um so yeah it's a real gift to be able to have these counterfactuals andy clark would argue that just qualitative experience is also another counterfactual so there will be a grandmother cell in your prefrontal cortex that fires when it says oh i've had the quality of experience of seeing red and it has to be like that in order for you to be able to be aware of it but using words like awareness now indeed self-awareness so quality of experience first of all um underpins self-awareness which would be the cultural experience of selfhood um so you know there are layers upon layers that you're talking about here each one of them has to be manifest in the genitive model so the first thing is that everything i've said so far um in terms of agency um could be applied to a thermostat so there's a thermostat conscious probably not so what kind of um system more complicated georgian model would you need to have in place before you could think yes possibly that could be conscious well the first one is a depth of planning so thermostat just does not look beyond the trajectories implied by the generalized motion and that's the gradients that are at hand you can't see half an hour to the future it just knows that your things are going up or going down in the moment very short term and um that sort of path um formulation or generative model but things like you and i can go and have much greater tempo depth and then really invert the notion of planning and then i repeat selecting amongst a number of different um counterfactual outcomes evaluating them and choosing one you get into issues of free will at this stage but still you can imagine writing a robot writing equipping a robot with these kinds of genetic models that people have done this um would you call that said right no then you have to have the hypothesis that it's you doing that and you in different different states doing that and then you have to say well why sorry i'm rushing because i i can see your actions to move what's the next question i just wanted to make it yeah it's a very special kind of generating model uh that even has selfhood let alone aware that you just have has suffered and even more rare to actually have um somebody who who worries about these about these things and not having the selfhood this is this is fascinating but you are a computationalist then you do believe that humans are independently conscious so first of all um i i think what you're saying if you are a computationalist is that if we could recreate all of the stuff that you're just talking about so forward reasoning and planning et cetera that we could recreate consciousness in silico yes yes i i see no reason why you couldn't create an artifact that was sufficiently similar to you and me um that would um you know the qualifier is very conscious yeah that i happen that's fascinating but yeah because the um the cyberneticists typically think that it's an emergent property and its computation is observer relative so it couldn't possibly exist in isolation and then that leads on to um assertions of pan psychism where consciousness must therefore be in everything um well can i ask for a clarification because yeah because i i think i'm i'm a computationalist as well so suppose um suppose we did create an in silico you know chip and some software that it had the depth of counter factual planning of temporal planning that could in fact evaluate its own actions and could even evaluate at a meta level you know the computations that are going into that that evaluation so that's sort of full stack almost the full recursive stack there would that thing be because a lot of times what i read from you know chalmers or other others is a question of feeling like what does it feel like to be conscious right and so would that thing in your view feel like something you know would it have the qualia of experience that we do yes um but you'd have to write it into the genitive model so you'd have to have to write the different feeling states as hypotheses about how am i feeling at the moment and then it would use or i i would use or you would use um all the messages and the belief updating and all the planning and all the um estimates of uncertainty that attend that planning and that's quite key the the the precision or the estimates of uncertainty start to take much much greater uh roles in terms of the heavy lifting of the brief updating the higher you get in the hierarchy so i am feeling in this way um i am aware of my cell phone i am having this quality of experience these would all be hypotheses um that were recognized in a good old-fashioned variation auto incurred away from everything that's going on below crucially including interceptive feelings of states of your battery your blood sugar and all of that good stuff um okay so i think if i understand you correctly and this is this is pretty cool so the feeling is actually a knowledge of a certain you know activity or trajectory like that is the feeling is in fact a knowledge absolutely well you could even say that the instance of the feeling the realization of the feeling is the inference in the context of a a knowledge structure that just is the generative model that has um representations of slots that have a semantics of i feel like a self i feel unhappy i feel anxious i feel in that life you know what whatever it is that is the best sense making the best hypothesis or explanation in you know a simple but not too simple way um for everything that's going on so feelings uh including things like pain and this is a you know this is quite an important so this is not just a philosophical nice nicety it really matters in the treatment um of chronic pain when you're identify the locus of pain and what is pain it seems that pain is a hypothesis that certain people bring to the table as the best explanation for all of their exceptional interceptive things fascinating i want to bring in the subject of um subjectivity because when we do try and formalize things and i know keith is smiling because we were just speaking with kenneth stanley and he has a wonderful book called why greatness cannot be planned and the myth of the objective he thinks that formalizing things actually blocks us from discovering interesting stepping stones in life and and um even in in the frame of our thinking it actually stops us from discovering new interesting knowledge and this is interesting in in your formalism so there is a chap called arthur eddington uh the notorious astronomer and physicist born in the in the 19th century and he spoke about subjectivity and science i think his point was that science only tells us a sliver of what's really happening in the world around us and we should be a lot less arrogant in our claims to understand it and kenneth stanley says that we've been too focused on objectives and the broader story is that he thinks society and institutions are scared of any subjectivity or perhaps you might think of it as um system one thinking so we always try to verbalize and understand everything but similarly to what you were just saying about in in respect of of agents and and the conscious experience and subjectivity it doesn't feel very objective anymore when you're talking about emotions yes um sorry um just the the measure of arthritis was was very pressing for me so that's how i got started this game with his book uh space driving gravitation my father made me recently that's it you know what's fascinating about that i got started in philosophy because i read his book the philosophy of physical sciences if only he did his family would be very proud to be in fact that his hand a century nearly i heard his now always essentially lead him absolutely but do you think there's a bit of an interesting contradiction there in yourself that as scientists we struggle to formalize everything we and you can't even get funding for a grant unless you have a clear hypothesis and a clear understanding of what you're doing and your papers have no value unless you have a clear formalism of what's going on um that was the spirit of the argument which i think i i certainly um uh concur with so i i don't see any uh real debate or dialectic here i mean this isn't simply a problem that we're talking about before in terms of the good scientist um being able to build um an apt hypothesis space where no one else has ever gone before and then do your your self-evidence your evidence-based signs but the clearly the the problem of building that hypothesis space you know that that that is that that's to my mind is unresolved your in the physical sciences um scoring any particular hypothesis absolutely no problem that's just estimating the marginal likelihood of the modern evidence or some some variation abound on it um and it can be you know it can be as difficult a hypothesis space as you as you want to consider but building it in the first place i think that's the uh that you know that is that is the outstanding problem but practically and philosophically and you know you have to ask are there any examples where it's been solved well yes naturals natural selection solved it and so it's used a particular way of exploring in a structured way using but say bisexual reproduction and that sort of gives you clues as to how you might um create and literally do lateral thinking or inventive thinking having ideas and hypotheses that no one ever has had before so um you know i i i think that that question if i read it properly just speaks to a really challenging exciting and practical challenge uh not just for sites but people trying to build sentient artifacts that now have in a principal way the ability to actually have new ideas and to test these new ideas out and i should just add that one example of these new ideas is that i am a person and i'm happy or in pain you know there's nothing i don't think they're particularly magical about uh the kind of constructs we talk about um in philosophy of selfhood unconsciousness and self-awareness and the psychophysics of visual perception it's just that you know having um these new kinds hypotheses props are not so new in terms of selfhood because of the culturalism that we're talking about and the fact mum talks about me and it's a very viable hypothesis that explains a lot and actually makes my data foraging on my epistemic uh foraging much more efficient just to simplify the world oh i am a person a non-busy person there are two cells that hypothesis perfectly fit for purpose maybe completely rubbish i don't know but it's a hypothesis that seems to work really well um and when you generalize it oh there are other persons sort of itself self self so i seem to be in the universe populate it myself this is a hypothesis which is reaffirmed through gathering the right kind of evidence through communication affording what caused by language and other aspects of it cultural psych evolutionists and psychology that again um speak to the sort of separation of temple skills but quite within this quite naturally selfhood arises um uh uh and one could argue uh philosophers also will flourish and we can have conversations about the heart problemation that the better heart problem as he will do but it all rests upon at some level at the highest level new hypotheses simpler better hypotheses being tested by some artifact or some markov blood kits the thing that fascinates me though is is that that with the um the fep formalism you were saying before about um self evidencing and finding our eco niche and when we do talk about well when we introduce subjectivity it doesn't feel it feels much more divergent and much more random and that's what kenneth stanley says he says that any any formalism is convergent and in order to be divergent we need to embrace subjectivity and it feels like emotions are the human form of subjectivity so that kind of in my in my mind that that makes this process feel much more random and divergent than perhaps the formalism would make us think again i i don't feel there's a i can't quite get a sense of dialect i don't know who disagree with that um it certainly is the case that finding your place and you know we actually demonstrate this in silico using cells that differentiate finding your place under a shared narrative or generative model uh amongst the population um is a free energy minimizing solution and it necessarily leverages or relies upon um subject-specific realizations that are contextualized by what the realizations another subject has so you can certainly um you can certainly argue that that subjectivity that subject-specific expression of um your sense making active sense making um is um a necessary part of the minimization of the joint free energy of a number of free energy minimizing systems even to the extent i mean this is um this is basically the holy grail and taken as read in things like computational psychiatry so yep if you want to use say the active inference um process theory to understand and phenotype people who might or may not have uh schizophrenia fall symptoms or may or may not show addictive behaviors or have this kind of autistic uh expression um then what you can do is you can use the complete class theorem to say well look in this game that i'm going to give you the complete classroom tells me that your behavior will be base optimal whatever you choose under some prize and what i want to know is what are your prize so that means for every person every subject there is a unique set of prize that characterize their behavior under ideal based assumptions that can be read as active inference so the job now is to try and identify what are these this what are this person's prize then use that to test hypotheses that you're this kind of person or that kind of person to try to find some systematic structure and um in the belief uh the belief structures that each subject brings to the table yeah before we run out of time i do want to ask you the the perennial free will question so you know i think and correct me if i'm wrong but i think you you believe in a universe that's unfolding according to these deterministic you know still albeit stochastic but deterministic you know flow equations um differential equations and so the universe is deterministic from that perspective and so in is that correct would you would you agree with sorry i did just technically um you you have random fluctuations um on on in a random dynamical system so you don't get deterministic case but you certainly get stochastic chaos uh absolutely and just interestingly uh it is becoming clearer that um things like you and me um may well have uh particular flows that are almost deterministic even if the outside is a bit random sure yeah and i would even i would even say it's entirely possible that those stochastic fluctuations are themselves highly chaotic you know deterministic systems or even if they are pure pure chance they're of the form of a chance that's you know not controllable um it's just you know it's just sort of this truly random chance and so in that context a lot of folks a lot of trouble imagining how we can have quote unquote free will from my perspective free will is simply that a an agent evaluates a set of inputs and sensory sensory things and has an algorithm that then determines you know what actions it's going to take you know based upon that and if you had put in different different sensory inputs it might have chosen a different different pathway like that's i'm talking about sort of degrees of freedom if you will as free will almost the same kind of concept i believe that that daniel bennett um espouses so i'm curious uh how you see free will in the context of a a universe unfolding either purely deterministically or deterministically with stochastic you know variation i think i say almost exactly the way that you you um you express that um just to make the point the the fast stochastic fluctuations of course um can also be um just um expressed at a faster time scale using their renormalization group so you don't even have to say i'm committed to stochastic as opposed to a deterministic chaos with a correlation dimension or greater than 10 for example you don't even have to do that once you allow for the separation of temporal scales and so you can talk about it you're a universe that is um it's no longer markovian because of the separation temple scales but certainly has the opportunity at one's given scale to have uh deterministic chaos and i think just in saying that um you're perfectly licensed to talk about free will in fact it would be difficult to imagine anything that plans and selects one of a number of ways forward you know those counterfactuals we talked about before that could not be construed as selecting and thereby and manifesting a piecing at least a minimal kind of willfulness um in the sense it has to do selection yeah what could argue well does that mean that natural selection is selecting you willfully does it make a choice does it have free will i'm not sure about that but but certainly if you if you if you have a generative model that sees itself doing its natural selection of the best counterfactuals about what to do or say next and then you know then there is uh that would i think be part of this sort of minimal selfhood um that you know would be undeniably a representation and an inference and a recognization of my free will and if i was able to infer that you were sufficiently like me i presumably had the same kind of jojo model i would also further you had free will and you would you know behave as such and i would secure that evidence by talking to you or just watching you and i would also leverage it i'd also use that uh as theory of mind to infer your intentional stance because i see you behaving in a way and if i behaved in that way i would have selected that plan and therefore i would have made that kind of choice and that would tell me the kind of person you are so it brings us back to the prize that characterize all of us in this instance if i can infer you very accurately we share the same prior commitments we have very very okay thank you amazing professor carl fristen thank you so much for joining us this afternoon it's been a pleasure oh i've enjoyed myself greatly thank you very much indeed wonderful